TUTORIAL 8

Adjugates and Vector Spaces

NOTATION AND CONVENTIONS

The set $F[X]$ consists of polynomials in X with coefficients in F, and $F[X]_{\leq d}$ consists of those of degree at most d, including zero. Equality means equality of coefficients at every power. Addition and scalar multiplication act on coefficients. The formal derivative of $p(X) = \sum_{j=0}^{d} a_j X^j$ is $p'(X) = \sum_{j=1}^{d} j a_j X^{j-1}$; over a finite field, j is interpreted as a field element. The supplementary appendix contains further polynomial material.

MOTIVATION AND GUIDING IDEA

Motivation. A homogeneous linear constraint allows two admissible states to be added, or an admissible state to be scaled, without leaving the admissible set. Span describes which states can be constructed; independence identifies components that are not redundant. Together, these ideas describe a system using a small collection of independent components.

NOTATION AND CONVENTIONS

Notation guide. $\mathbb{F}$ is the scalar field, V a vector space over it, 0_V the zero vector, and $\alpha \in \mathbb{F}$ a scalar. $v \in S$ denotes membership and $S \subseteq V$ inclusion. $\operatorname{span}_{\mathbb{F}} S$ is the span over the indicated field. $U \cup W$ denotes union, $U \cap W$ intersection, $U + W$ a sum of subspaces, and $U \oplus W$ a direct sum. The symbol $\setminus$ denotes set difference and $\varnothing$ the empty set.

DEFINITION 8.1

Vector space axioms and solution tools. A vector space over a field F is a set V equipped with operations $V \times V \to V$ and $F \times V \to V$ satisfying, for all $u, v, w \in V$ and $\alpha, \beta \in F$,

$$u + v = v + u, \qquad (u + v) + w = u + (v + w).$$

There is $0_V \in V$ such that $v + 0_V = v$, and every $v \in V$ has an additive inverse $-v \in V$ satisfying $v + (-v) = 0_V$. In addition,

$$\alpha(u + v) = \alpha u + \alpha v, \qquad (\alpha + \beta)v = \alpha v + \beta v,$$

$$(\alpha\beta)v = \alpha(\beta v), \qquad 1_F v = v.$$

For $A \in M_{m\times n}(F)$, the system $Ax = b$ is consistent if and only if $\operatorname{rank} A = \operatorname{rank}[A \mid b]$. If x_0 is a particular solution, then its full solution set is $x_0 + \ker A$: indeed, $Ax = b$ is equivalent to $A(x - x_0) = 0$. This solution set is a subspace of F^n if and only if $b = 0$, since containing the zero vector forces $A0 = b = 0$.

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NOTE

A vector is an element of the vector space, so a matrix or polynomial can also be a vector. In an independence test, the unknowns are the *coefficients of the combination*; in a span-membership test, the target vector is given. For a nonhomogeneous condition such as $x + 2y + z = 3$, first check whether the zero vector belongs to the set.

MOTIVATION AND GUIDING IDEA

Objectives. We will formulate problems as linear systems, perform row reduction while preserving the solution set, and distinguish free variables from pivot variables. We will connect homogeneous systems with vector spaces and test closure systematically.

MOTIVATION AND GUIDING IDEA

Tutorial objectives. Test linear dependence and independence, find spanning sets, and distinguish the union, intersection, and sum of subspaces. State the scalar field in every test and connect the equations with their geometric interpretation.

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8.1 The Classical Adjugate

DEFINITION 8.2

Let $A = (a_{ij}) \in M_n(\mathbb{F})$ with $n \geq 2$. Write $A_{\widehat{i,j}}$ for the matrix obtained by deleting row i and column j. The scalar

$$C_{ij} = (-1)^{i+j} \det A_{\widehat{i,j}}$$

is its (i, j) cofactor. The classical adjugate is

$$\operatorname{adj}(A) = (C_{ij})^T.$$

For $n = 1$, set $\operatorname{adj}((a)) = (1)$. The notation $\operatorname{adj}(A)$ denotes the classical adjugate. The Hermitian adjoint will be denoted by $A^* = \overline{A}^T$.

PROPOSITION 8.1

Every square matrix over a field satisfies

$$A \operatorname{adj}(A) = \operatorname{adj}(A)A = (\det A)I_n.$$

Indeed, entry (i, j) of the first product is

$$(A \operatorname{adj}(A))_{ij} = \sum_{k=1}^{n} a_{ik}C_{jk}.$$

For $i = j$, this is expansion along row j. For $i \neq j$, it is the determinant expansion of the matrix obtained by replacing row j with row i. That matrix has two identical rows and determinant zero. Expansion along columns proves the identity for the second product. In

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particular, if $\det A \neq 0$, then

$$A^{-1} = \frac{1}{\det A}\,\mathrm{adj}(A).$$

EXAMPLE 8.1

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$,

$$(C_{ij}) = \begin{pmatrix} d & -c \\ -b & a \end{pmatrix}, \qquad \mathrm{adj}(A) = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Both products can be computed directly:

$$A\,\mathrm{adj}(A) = \begin{pmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{pmatrix} = (ad - bc)I_2,$$

$$\mathrm{adj}(A)A = \begin{pmatrix} da - bc & db - bd \\ -ca + ac & -cb + ad \end{pmatrix} = (ad - bc)I_2.$$

When $ad - bc \neq 0$, division by that scalar gives the usual inverse formula.

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EXERCISE 8.1

Consider the real matrix

$$A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 2 \end{pmatrix}.$$

Please compute $\det A$, every cofactor, $\operatorname{adj}(A)$, and A^{-1}. Please verify both adjugate product identities.

WORKED SOLUTION

The matrix is triangular, so $\det A = 1 \cdot 1 \cdot 2 = 2$. The minors and signs give

$$\begin{aligned} C_{11} &= 1 \cdot 2 - 3 \cdot 0 = 2, & C_{12} &= -(0 \cdot 2 - 3 \cdot 0) = 0, & C_{13} &= 0 \cdot 0 - 1 \cdot 0 = 0, \\ C_{21} &= -(2 \cdot 2 - 0 \cdot 0) = -4, & C_{22} &= 1 \cdot 2 - 0 \cdot 0 = 2, & C_{23} &= -(1 \cdot 0 - 2 \cdot 0) = 0, \\ C_{31} &= 2 \cdot 3 - 0 \cdot 1 = 6, & C_{32} &= -(1 \cdot 3 - 0 \cdot 0) = -3, & C_{33} &= 1 \cdot 1 - 2 \cdot 0 = 1. \end{aligned}$$

Thus

$$\operatorname{adj}(A) = \begin{pmatrix} 2 & -4 & 6 \\ 0 & 2 & -3 \\ 0 & 0 & 1 \end{pmatrix}, \qquad A^{-1} = \begin{pmatrix} 1 & -2 & 3 \\ 0 & 1 & -3/2 \\ 0 & 0 & 1/2 \end{pmatrix}.$$

Direct multiplication verifies

$$A \operatorname{adj}(A) = \begin{pmatrix} 2 & -4+4 & 6-6 \\ 0 & 2 & -3+3 \\ 0 & 0 & 2 \end{pmatrix} = 2I_3,$$

$$\operatorname{adj}(A) A = \begin{pmatrix} 2 & 4-4 & -12+12 \\ 0 & 2 & 6-6 \\ 0 & 0 & 2 \end{pmatrix} = 2I_3.$$

Dividing by 2 also verifies $AA^{-1} = A^{-1}A = I_3$.

We've got it. Victory!

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8.2 Vector spaces

RECALL AND TOOLS

1. To define a vector space V over a field $\mathbb{F}$, we specify an addition of vectors in V and a multiplication of a vector in V by a scalar in $\mathbb{F}$, satisfying the vector space axioms. These are **in addition** to addition and multiplication in $\mathbb{F}$. A product of two vectors is not part of the vector space axioms; if needed later, it will be defined separately.
2. Every field is a vector space over itself, with the field's addition and multiplication.
3. A vector space V is always defined over a field $\mathbb{F}$. Always specify the field of scalars. For example, as discussed in the lecture, $\mathbb{C}$ can be regarded as a vector space over $\mathbb{C}$ or over $\mathbb{R}$. These are two different vector spaces.

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EXAMPLE 8.2

Let $\mathbb{F}$ be a field.

(a) $\mathbb{F}^n$: the set of n-tuples over $\mathbb{F}$, where $n \in \mathbb{N}$.

(b) $M_{m\times n}(\mathbb{F})$: the set of $m \times n$ matrices over $\mathbb{F}$.

(c) $\mathbb{F}[X]$: the set of all polynomials with coefficients in $\mathbb{F}$.

EXERCISE 8.2

Please determine whether the following sets are subspaces of $\mathbb{R}[X]$ over $\mathbb{R}$:

1. $V = \{p(X) \in \mathbb{R}[X] : p(-2) = 0,\ p(3) \neq 0\}$.
2. $V = \{p(X) \in \mathbb{R}[X] : \exists c \in \mathbb{R} \setminus \{0\},\ p(c) = 0\}$.

WORKED SOLUTION

1. No. The zero polynomial is absent. Addition also fails: for
$$p(X) = (X+2)(X-8), \qquad q(X) = (X+2)^2,$$
we have $p(-2) = q(-2) = 0$ and $p(3) = -25$, $q(3) = 25$, but
$$(p+q)(X) = (X+2)(2X-6), \qquad (p+q)(3) = 0.$$
2. No. The polynomials $(X-2)^2$ and $(X+2)^2$ belong to the set, with roots $2, -2$, respectively. Their sum is $2X^2 + 8 > 0$ for all real X, so it does not belong.

That does it. Victory!

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8.3 Subspaces

DEFINITION 8.3

Let V be a vector space over a field $\mathbb{F}$, and let $U \subseteq V$ be a subset that is itself a vector space over $\mathbb{F}$ under the addition and scalar multiplication inherited from V. Then U is a vector subspace of V.

PROPOSITION 8.2

Let V be a vector space over a field $\mathbb{F}$, and let $U \subseteq V$ be a subset. Then U is a vector subspace if and only if all of the following conditions hold:

1. $0 \in U$, where this is the zero vector of V.
2. U is closed under addition.
3. U is closed under scalar multiplication.

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EXAMPLE 8.3

1. The set $V = \{p(X) \in F[X] : p(-3) = 0\}$ is a subspace of $F[X]$. It contains the zero polynomial. For $p, q \in V$ and $\alpha \in F$,

$$(p+q)(-3) = p(-3) + q(-3) = 0, \qquad (\alpha p)(-3) = \alpha p(-3) = 0.$$

   Thus it is closed under addition and scalar multiplication.

2. For $n \geq 3$, the set $W = \{x \in \mathbb{R}^n : x_1 + x_3 = 0\}$ is a subspace. It contains zero. If $x, y \in W$, then

$$(x_1 + y_1) + (x_3 + y_3) = (x_1 + x_3) + (y_1 + y_3) = 0.$$

   Also $(\alpha x)_1 + (\alpha x)_3 = \alpha(x_1 + x_3) = 0$ for every $\alpha \in \mathbb{R}$.

EXERCISE 8.3

Please prove or please disprove:

1. The set $U = \{(a, 3a^2) : a \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^2$.
2. The symmetric matrices form a subspace of $M_n(F)$.

WORKED SOLUTION

1. This is not a subspace: $(1, 3) \in U$, but

$$(1,3) + (1,3) = (2,6) \notin U,$$

   because a first coordinate of 2 requires second coordinate $3 \cdot 2^2 = 12$.

2. The zero matrix is symmetric. If $A^T = A$, $B^T = B$ and $\lambda \in F$, then

$$(A+B)^T = A^T + B^T = A + B, \qquad (\lambda A)^T = \lambda A^T = \lambda A.$$

   Thus all three subspace-test conditions hold.

All wrapped up. Victory!

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8.4 Vector Spaces

RECALL AND TOOLS

1. To define a vector space V over a field $\mathbb{F}$, specify addition of vectors in V and multiplication of a vector in V by a scalar in $\mathbb{F}$, satisfying the vector-space axioms. These operations are specified **in addition** to the addition and multiplication in $\mathbb{F}$. Multiplication of two vectors is not a vector-space axiom; any such operation used later must be defined separately.
2. Every field is a vector space over itself, using its own addition and multiplication.

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3. A vector space V is always defined over a field $\mathbb{F}$, which must be stated. For example, as discussed in the lecture, $\mathbb{C}$ can be viewed as a vector space over $\mathbb{C}$ or over $\mathbb{R}$. These are two different vector-space structures.

EXAMPLE 8.4

Let $\mathbb{F}$ be a field.

(a) $\mathbb{F}^n$: the set of n-tuples over $\mathbb{F}$ ($n \in \mathbb{N}$).

(b) $M_{m\times n}(\mathbb{F})$: the set of $m \times n$ matrices over $\mathbb{F}$.

(c) $\mathbb{F}[X]$: the set of all polynomials with coefficients in $\mathbb{F}$.

EXERCISE 8.4

Please determine whether the following sets are subspaces of $\mathbb{R}[X]$ over $\mathbb{R}$ with the usual operations.

1. $V = \{p(X) \in \mathbb{R}[X] : p(-3) = 0,\ p(2) \neq 0\}$.
2. $W = \{p(X) \in \mathbb{R}[X] : \exists c \in [0, 1],\ p(c) = 0\}$.

WORKED SOLUTION

1. No. The zero polynomial does not belong to V. Addition also fails: for
$$p(X) = (X + 3)(X - 1), \qquad q(X) = -(X + 3),$$
we have $p(-3) = q(-3) = 0$, $p(2) = 5$, and $q(2) = -5$. Thus $p, q \in V$, but
$$(p + q)(X) = (X + 3)(X - 2), \qquad (p + q)(2) = 0,$$
so $p + q \notin V$.
2. No. Both $p(X) = X$ and $q(X) = 1 - X$ belong to W, with roots 0 and 1. Their sum is the constant polynomial 1, which has no root in the interval. Thus W is not closed under addition.

We've reached the finish. Victory!

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8.4.1 Subspaces

DEFINITION 8.4

Let V be a vector space over a field $\mathbb{F}$. A subset $U \subseteq V$ is a **vector subspace** of V if it is itself a vector space over $\mathbb{F}$ under the addition and scalar multiplication inherited from V.

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PROPOSITION 8.3

Let V be a vector space over a field $\mathbb{F}$ and $U \subseteq V$. Then U is a subspace if and only if all the following conditions hold:

1. $0 \in U$, where 0 is the zero vector of V.
2. U is closed under addition.
3. U is closed under scalar multiplication.

EXAMPLE 8.5

1. $V = \{p(X) \in \mathbb{F}[X] : p(-3) = 0\}$.
2. $U = \{x \in \mathbb{R}^n : x_1 = 2x_2\}$, where $n \geq 2$.

Both sets are subspaces. The zero polynomial lies in V, and for $p, q \in V$ and $\alpha \in \mathbb{F}$,

$$(p+q)(-3) = p(-3) + q(-3) = 0, \qquad (\alpha p)(-3) = \alpha p(-3) = 0.$$

The zero vector also lies in U. If $x, y \in U$ and $\alpha \in \mathbb{R}$, then

$$(x+y)_1 = x_1 + y_1 = 2x_2 + 2y_2 = 2(x+y)_2,$$

$$(\alpha x)_1 = \alpha x_1 = 2\alpha x_2 = 2(\alpha x)_2.$$

Thus both sets are closed under addition and scalar multiplication.

EXERCISE 8.5

Please prove or disprove.

1. $U = \{(a, 5a^2)^T : a \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^2$.
2. The symmetric matrices form a subspace of $M_n(\mathbb{F})$.

WORKED SOLUTION

1. No.
$$(1,5)^T \in U, \qquad (1,5)^T + (1,5)^T = (2,10)^T \notin U,$$
because $5 \cdot 2^2 = 20 \neq 10$, so addition is not closed.
2. Yes. $0^T = 0$, and if $A^T = A$, $B^T = B$, and $\lambda \in \mathbb{F}$, then
$$(A+B)^T = A^T + B^T = A + B, \qquad (\lambda A)^T = \lambda A^T = \lambda A.$$
All three subspace conditions hold.

Yes! Victory!

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8.5 Linear Combinations

DEFINITION 8.5

Let V be a vector space over a field $\mathbb{F}$ and let $v_1, \ldots, v_k \in V$. A vector $v \in V$ is a **linear combination** of $v_1, \ldots, v_k$ if there are $a_1, \ldots, a_k \in \mathbb{F}$ such that $v = a_1 v_1 + \cdots + a_k v_k$.

DEFINITION 8.6

Let V be a vector space over a field $\mathbb{F}$. For a set $S \subseteq V$, not necessarily finite, define $\operatorname{span}_{\mathbb{F}} S$ as the set of all finite linear combinations of elements of S:

$$\operatorname{span}_{\mathbb{F}} S = \left\{ \sum_{j=1}^{k} \alpha_j v_j \;\middle|\; k \geq 0,\ v_j \in S,\ \alpha_j \in \mathbb{F} \right\}.$$

The empty sum ($k = 0$) is allowed and equals the zero vector. In particular, $\operatorname{span}_{\mathbb{F}} \varnothing = \{0\}$.

NOTE

If V is a vector space over $\mathbb{F}$ and $S \subseteq V$, then $\operatorname{span} S$ is a subspace of V.

DEFINITION 8.7

If V is a vector space over $\mathbb{F}$ and $S \subseteq V$ satisfies $V = \operatorname{span} S$, then S **spans** V and is called a **spanning set**. If S is finite, V is **finitely generated**.

NOTE

Pay attention to the scalar field. A set $S \subseteq V$ can span over one field and fail to span over another. When more than one field is possible, write $\operatorname{span}_{\mathbb{F}} S$ to specify the field.

EXERCISE 8.6

Let V be a vector space over $\mathbb{F}$, let $\varnothing \neq S \subseteq V$ be a finite set of size k, and let $v \in V$. Please prove that

$$\operatorname{span}(S \cup \{v\}) = \operatorname{span} S \quad \Longleftrightarrow \quad v \in \operatorname{span} S.$$

WORKED SOLUTION

$\Longrightarrow$: Assume $\operatorname{span}(S \cup \{v\}) = \operatorname{span} S$. Write $v = 0 \cdot v_1 + \cdots + 0 \cdot v_k + 1 \cdot v$. Thus v is a linear combination of elements of $S \cup \{v\}$, so $v \in \operatorname{span}(S \cup \{v\})$. The assumed equality gives $v \in \operatorname{span} S$.

$\Longleftarrow$: The inclusion $\operatorname{span}(S \cup \{v\}) \supseteq \operatorname{span} S$ always holds: any linear combination of elements of S is also one of elements of $S \cup \{v\}$, with coefficient zero on v. For the opposite inclusion, write $S = \{s_1, \ldots, s_k\}$ and take $x \in \operatorname{span}(S \cup \{v\})$. There are $a_1, \ldots, a_{k+1} \in \mathbb{F}$ with

$$a_1 s_1 + \cdots + a_k s_k + a_{k+1} v = x.$$

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EXERCISE 8.6 CONTINUED

Since $v \in \operatorname{span} S$, there are $b_1, \ldots, b_k \in \mathbb{F}$ with $b_1 s_1 + \cdots + b_k s_k = v$. Substituting gives

$$(a_1 + a_{k+1} b_1) s_1 + \cdots + (a_k + a_{k+1} b_k) s_k = x.$$

Hence $x \in \operatorname{span} S$.

V-I-C-T-O-R-Y!

DEFINITION 8.8

Let V be a vector space over $\mathbb{F}$ and $S \subseteq V$.

1. The set S is **linearly dependent** if there are pairwise distinct vectors $v_1, \ldots, v_k \in S$ and scalars $\alpha_1, \ldots, \alpha_k \in \mathbb{F}$, not all zero, such that
$$\alpha_1 v_1 + \cdots + \alpha_k v_k = 0.$$
2. The set S is **linearly independent** if, for every finite choice of pairwise distinct vectors $v_1, \ldots, v_k \in S$, this equality implies $\alpha_1 = \cdots = \alpha_k = 0$.

The vectors need not be tuples: they may also be polynomials, matrices, or elements of another vector space.

8.6 Bases and Dimension

DEFINITION 8.9

Let V be a vector space over a field $\mathbb{F}$. A set $\{u_1, \ldots, u_n\} \subseteq V$ is a **basis** of V if it is linearly independent and spans V.

A finite-dimensional vector space can have many bases, but all its bases have the same number of elements. This number is the **dimension** of the space.

EXAMPLE 8.6

$$\dim(\mathbb{F}^n) = n, \qquad \dim M_{m\times n}(\mathbb{F}) = mn, \qquad \dim \mathbb{F}_d[x] = d + 1.$$

For example, the standard basis of $\mathbb{R}^3$ is

$$e_1 = (1,0,0)^T;\ e_2 = (0,1,0)^T;\ e_3 = (0,0,1)^T.$$

The standard basis of $\mathbb{R}_3[x]$ is $(1, x, x^2, x^3)$.

RECALL AND TOOLS

In a vector space of known finite dimension n, any two of the following three properties suffice to establish that a set is a basis:

1. Linear independence.
2. Spanning the space.

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3. Having exactly n elements, the dimension of the space.

8.7 Bases and Dimension

DEFINITION 8.10

Let V be a vector space over a field $\mathbb{F}$. A set $\{u_1, \ldots, u_n\} \subseteq V$ is a **basis** of V if it is linearly independent and spans V.

A finite-dimensional vector space can have many bases, but all its bases have the same number of elements. This number is the **dimension** of the space.

EXAMPLE 8.7

$$\dim(\mathbb{F}^n) = n, \qquad \dim M_{m\times n}(\mathbb{F}) = mn, \qquad \dim \mathbb{F}_d[x] = d+1.$$

For example, the standard basis of $\mathbb{R}^4$ is

$$e_1 = (1,0,0,0)^T;\ e_2 = (0,1,0,0)^T;\ e_3 = (0,0,1,0)^T;\ e_4 = (0,0,0,1)^T.$$

The standard basis of $\mathbb{R}_1[x]$ is $(1, x)$.

RECALL AND TOOLS

Let V be a vector space over a field $\mathbb{F}$ with finite dimension, $\dim V < \infty$.

1. If B_1, B_2 are bases of V, then $|B_1| = |B_2|$. Thus dimension is well-defined: $\dim V$ is the cardinality of any basis of V.
2. If $W = \{w_1, \ldots, w_m\} \subseteq V$ is linearly independent, then
$$|W| = m \leq \dim V.$$
One cannot choose more independent vectors than the number of vectors in a basis.
3. If $W = \{w_1, \ldots, w_m\} \subseteq V$ spans V, that is, $V = \operatorname{span}(W)$, then
$$|W| = m \geq \dim V.$$
Every spanning set therefore has at least $\dim V$ vectors.
4. Suppose $W = \{w_1, \ldots, w_m\} \subseteq V$ satisfies:
   - $\dim V = |W| = m$;
   - W is either linearly independent or spans V.

   Then W is a basis of V. If W is independent and has exactly $\dim V$ vectors, no further vector can be added while preserving independence, so W already spans. If W spans and has exactly $\dim V$ vectors, there can be no nontrivial zero linear combination of its elements, so they are independent.

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PROPOSITION 8.4

An additional important fact is the following. Let V have dimension n, and let $W = \{w_1, \ldots, w_k\} \subseteq V$ be linearly independent, with $k \leq n$. There is a set

$$U = \{u_{k+1}, \ldots, u_n\} \subseteq V$$

such that

$$B = W \cup U = \{w_1, \ldots, w_k, u_{k+1}, \ldots, u_n\}$$

is a basis of V. Thus **every linearly independent set in a finite-dimensional space can be extended to a basis**. This is a central tool for constructing bases and calculating dimensions.

RECALL AND TOOLS

To establish that a set is a basis of a space of known finite dimension n, it suffices to verify any two of:

1. Linear independence.
2. Spanning the space.
3. Having exactly n elements, the dimension of the space.

8.8 The Dimension Formula

PROPOSITION 8.5

Let V be a finitely generated vector space over a field $\mathbb{F}$, and let U be a subspace of V.

1. U is finitely generated and $\dim U \leq \dim V$.
2. $\dim U = \dim V$ if and only if $U = V$.

8.9 The Dimension Formula

PROPOSITION 8.6

Let V be a finitely generated vector space over a field $\mathbb{F}$, and let U be a subspace of V.

1. U is finitely generated and $\dim U \leq \dim V$.
2. $\dim U = \dim V$ if and only if $U = V$.

EXERCISE 8.7

Let $p(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$ be a real polynomial.

1. Please find all polynomials satisfying $p(0) = 0$, $p(2) = 4$, $p'(0) = 0$, $p'(1) = 0$. Please determine whether the solution set is a subspace and whether $t = 1$ is a local maximum or minimum.

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EXERCISE 8.7 CONTINUED

2. Please repeat with $p(0) = 2$, $p(2) = 6$, $p'(0) = 0$, $p'(1) = 0$.

3. Add $p'(2) = 5$ to the second part. Please determine whether the system is consistent and whether its solution set is a subspace.

WORKED SOLUTION

We have $p'(t) = a_1 + 2a_2t + 3a_3t^2$. In part one, $a_0 = a_1 = 0$, while the remaining conditions give

$$4a_2 + 8a_3 = 4, \qquad 2a_2 + 3a_3 = 0.$$

Subtracting twice the second equation from the first gives $2a_3 = 4$, so $a_3 = 2$ and $a_2 = -3$. The unique solution is

$$p_1(t) = -3t^2 + 2t^3.$$

Check: $p_1(2) = -12 + 16 = 4$ and $p_1'(1) = -6 + 6 = 0$. The zero polynomial fails $p(2) = 4$, so the solution set is not a subspace. Also $p_1''(t) = -6 + 12t$, hence $p_1''(1) = 6 > 0$, giving a local minimum.
In part two, $a_0 = 2$, $a_1 = 0$, and the other equations are unchanged. Thus

$$p_2(t) = 2 - 3t^2 + 2t^3.$$

Again the solution is unique, its set is not a subspace, and $t = 1$ is a local minimum. In part three, this unique polynomial gives

$$p_2'(2) = -12 + 24 = 12 \neq 5.$$

The system is inconsistent, and the empty solution set is not a subspace.

🏆 **Victory!**

EXERCISE 8.8

Let

$$p(t) = a_0 + a_1t + a_2t^2 + a_3t^3 \in \mathbb{R}_3[t].$$

1. Please find all polynomials satisfying $p(0) = 0$, $p(1) = -1$, and $p'(0) = p'(1/2) = 0$. Please determine whether the solution set is a subspace and whether $t = 1/2$ is a local maximum or minimum.

2. Please repeat for $p(0) = 2$, $p(1) = 1$, and $p'(0) = p'(1/2) = 0$.

3. Please add $p'(1) = 1$ to the preceding conditions and please determine whether a solution exists.

WORKED SOLUTION

$$p'(t) = a_1 + 2a_2t + 3a_3t^2.$$

In part 1, $a_0 = a_1 = 0$. The remaining conditions give

$$a_2 + a_3 = -1, \qquad a_2 + \tfrac{3}{4}a_3 = 0.$$

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EXERCISE 8.8 CONTINUED

Subtracting gives $a_3/4 = -1$, so $a_3 = -4$ and $a_2 = 3$. The unique solution is

$$p_1(t) = 3t^2 - 4t^3.$$

Indeed, $p_1(1) = -1$ and $p_1'(1/2) = 3 - 3 = 0$. The solution set is not a subspace because the zero polynomial fails $p(1) = -1$. Also,

$$p_1''(t) = 6 - 24t, \qquad p_1''(1/2) = -6 < 0,$$

so this is a local maximum.
In part 2, $a_0 = 2$, $a_1 = 0$, and $a_2 + a_3 = 1 - 2 = -1$. The same system determines a_2, a_3, giving

$$p_2(t) = 2 + 3t^2 - 4t^3.$$

The solution is again unique, and its set is not a subspace because it does not contain the zero polynomial. The derivatives coincide, so $t = 1/2$ is again a local maximum.
In part 3, $p_2'(1) = 6 - 12 = -6 \neq 1$. Thus the additional condition contradicts the preceding ones. The solution set is empty and is not a subspace.

And we're done. Victory!

DEFINITION 8.11

Hypotheses and tools. Let V have dimension $n < \infty$ over $\mathbb{F}$. Every independent set in V can be extended to a basis; relative to an ordered basis, every $v \in V$ has a unique expression. If $A \in M_n(\mathbb{F})$, $n \geq 1$, is invertible, then $A^{-1} \in M_n(\mathbb{F})$ satisfies $AA^{-1} = A^{-1}A = I_n$. The inverse is unique: if B, C are inverses, then $B = B(AC) = (BA)C = C$.
The following are equivalent: A is invertible; its reduced row-echelon form is I_n; rank $A = n$; $Ax = b$ has a unique solution for every $b \in \mathbb{F}^n$; $Ax = 0$ has only the zero solution; the columns of A form a basis of $\mathbb{F}^n$. For square matrices of the same order, $(AB)^{-1} = B^{-1}A^{-1}$ when both factors are invertible.
If row operations are applied in the order $E_1, E_2, \ldots, E_k$, the result is $E_k \cdots E_2 E_1 A$. If this equals I_n, then $A^{-1} = E_k \cdots E_1$, whereas $A = E_1^{-1} \cdots E_k^{-1}$. The factor order is part of the assertion.

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Additional practice

8.10 Supplementary Exercises: Span, Redundancy, and the Scalar Field

EXERCISE 8.9

$$U = \{(x, y, z) \in \mathbb{R}^3 : x + 2y + z = 0\}, \qquad u = (2, -1, 0), \quad v = (0, 1, -2), \quad w = (2, 0, -2).$$

Please prove $U = \text{span}\{u, v\}$. Please test independence and the effect of adding w. Please determine whether $x + 2y + z = 3$ defines a subspace.

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EXERCISE 8.9 CONTINUED

WORKED SOLUTION

$$au + bv = (2a, -a + b, -2b), \qquad 2a + 2(-a + b) - 2b = 0.$$

Thus $\text{span}\{u, v\} \subseteq U$. Conversely, $(x, y, z) \in U$ implies $y = -(x + z)/2$, so

$$(x, y, z) = \frac{x}{2}u - \frac{z}{2}v.$$

Hence $U = \text{span}\{u, v\}$. In $au + bv = 0$, the first and third coordinates give $a = b = 0$, proving independence. Since $w = u + v$, adding w preserves the span but introduces the dependence $u + v - w = 0$. The set defined by $x + 2y + z = 3$ is not a subspace because it excludes the zero vector.

You know what that means? V-I-C-T-O-R-Y!

EXERCISE 8.10

For $S = \{1, 2 + i\} \subseteq \mathbb{C}$, please test linear dependence and spanning over $\mathbb{R}$ and over $\mathbb{C}$.

WORKED SOLUTION

Over $\mathbb{R}$, $a + b(2 + i) = 0$ gives $b = 0$ from the imaginary part, then $a = 0$. Thus the pair is independent. Every $x + iy \in \mathbb{C}$ satisfies

$$x + iy = (x - 2y) \cdot 1 + y(2 + i),$$

so the pair spans $\mathbb{C}$ over $\mathbb{R}$. Over $\mathbb{C}$, the nontrivial relation

$$-(2 + i) \cdot 1 + 1 \cdot (2 + i) = 0$$

shows dependence. The pair still spans because 1 alone spans $\mathbb{C}$ over itself.

Another one down. Victory!

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EXERCISE 8.11 CHALLENGE

Let $S \subseteq \mathbb{R}^n$ be nonempty and assume $(1-t)x + ty \in S$ for all $x, y \in S$ and $t \in \mathbb{R}$.

1. Please choose $a \in S$ and prove that $U = \{x - a : x \in S\}$ is a subspace and $S = a + U$, where $a + U = \{a + u : u \in U\}$.
2. Is it enough to require only $(x + y)/2 \in S$ for all $x, y \in S$? Please prove or disprove.

WORKED SOLUTION

We have $0 = a - a \in U$. For $u \in U$ and $\lambda \in \mathbb{R}$, the hypothesis gives

$$(1 - \lambda)a + \lambda(a + u) = a + \lambda u \in S,$$

so $\lambda u \in U$. For $u, v \in U$, taking the midpoint of $a + u$ and $a + v$ gives $(u + v)/2 \in U$. Scalar closure with scalar 2 then gives $u + v \in U$. Hence U is a subspace. Its definition gives both inclusions in $S = a + U$. Midpoint closure alone is insufficient. The set $\mathbb{Q} \subseteq \mathbb{R}$ is closed under midpoints. If it were $a + U$ for a real subspace U, the fact that $0 \in \mathbb{Q}$ would imply $\mathbb{Q} = U$. But $1 \in U$ would then imply $\sqrt{3} \cdot 1 \in U$, a contradiction.

We've got it. Victory!

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EXERCISE 8.12 CHALLENGE

Let $U_1, \ldots, U_m$ be proper subspaces of a real vector space V, meaning $U_j \neq V$ for every j.

1. Please prove that $U_1 \cup \cdots \cup U_m \neq V$.
2. Please construct four proper subspaces of $\mathbb{Z}_3^2$ whose union is the whole space.

WORKED SOLUTION

We induct on m. The case $m = 1$ is the assumption. By the induction hypothesis, choose

$$a \notin U_1 \cup \cdots \cup U_{m-1}, \qquad b \notin U_m.$$

Consider $a + tb$ for $t \in \mathbb{R}$. For each $j < m$, at most one value of t gives a point in U_j. Two distinct values would imply $b \in U_j$ by subtraction and division by their difference, and then $a \in U_j$, a contradiction. The same line meets U_m in at most one point, since two would imply $b \in U_m$. Only finitely many parameters are excluded, whereas $\mathbb{R}$ is infinite. This proves the claim. Over $\mathbb{Z}_3$, take

$$U_1 = \operatorname{span}\{(1,0)\}, \quad U_2 = \operatorname{span}\{(0,1)\},$$
$$U_3 = \operatorname{span}\{(1,1)\}, \quad U_4 = \operatorname{span}\{(1,2)\}.$$

Each subspace has three vectors, so each is proper. A vector (x, y) with $x = 0$ belongs to U_2. If $x \neq 0$, then $(x, y) = x(1, y/x)$, where y/x is 0, 1, or 2, so the vector lies in U_1, U_3, or U_4. The union covers all nine vectors.

That does it. Victory!

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